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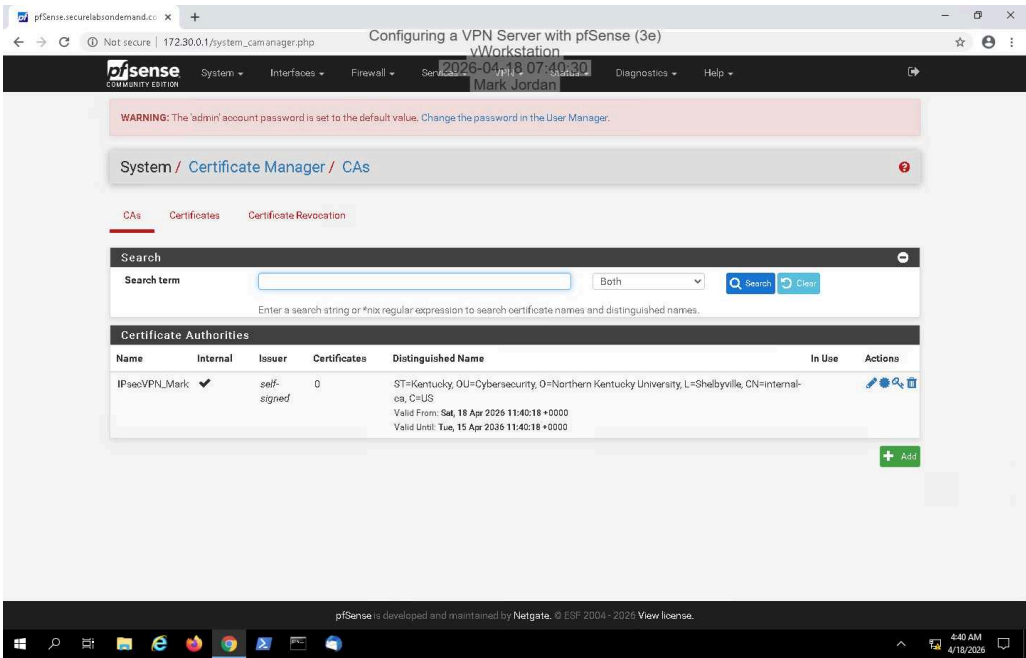
Time on Task:	Progress:
1 hour, 6 minutes	100%

Report Generated: Saturday, April 18, 2026 at 8:40 AM

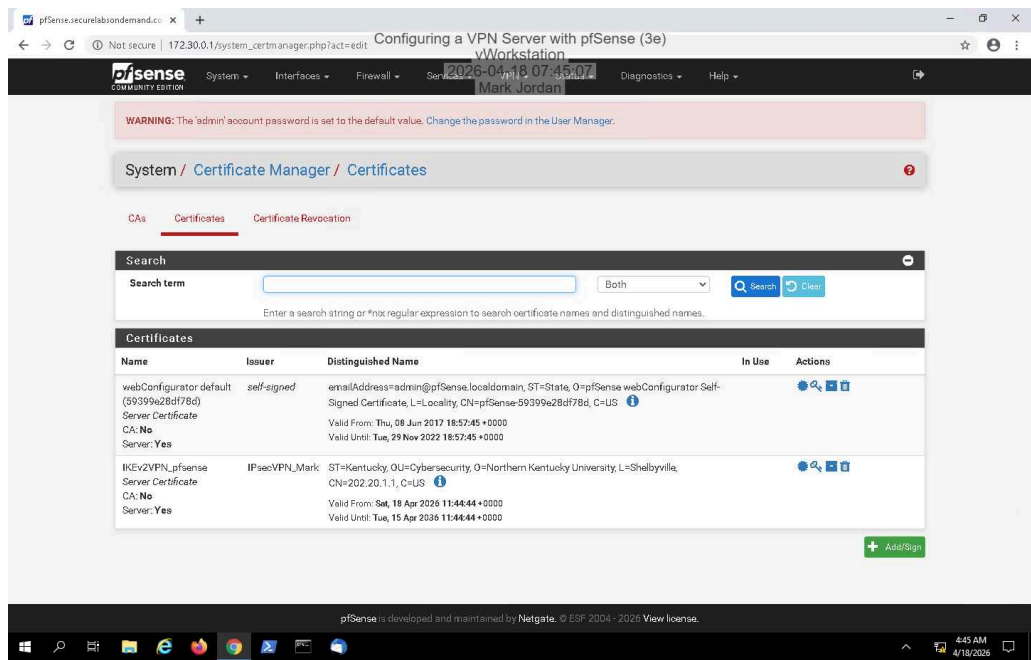
Section 1: Hands-On Demonstration

Part 1: Configure an IPsec VPN Server

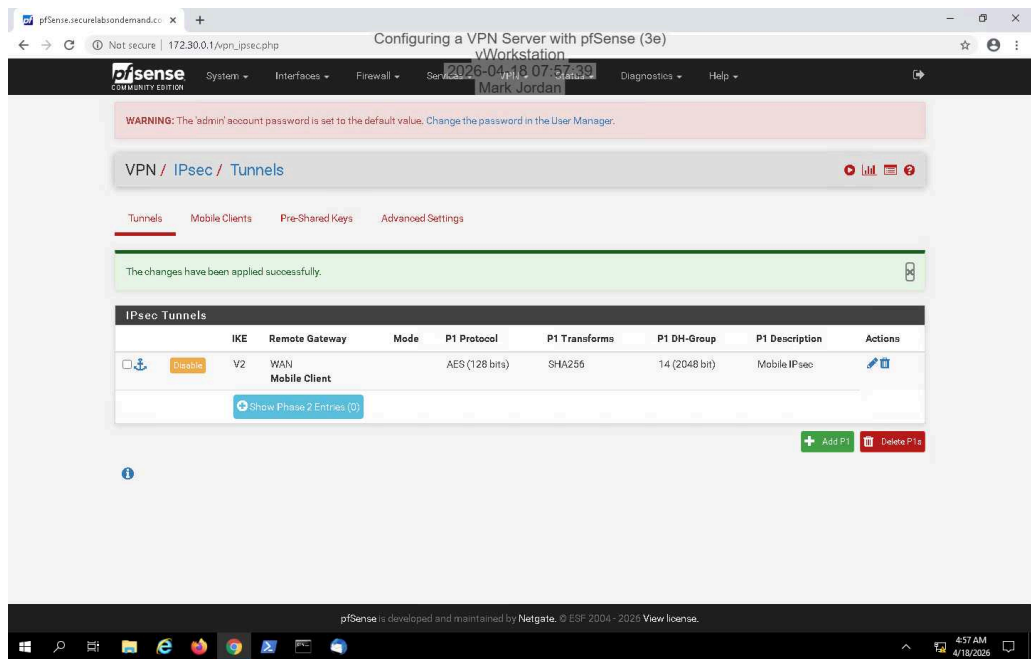
18. Make a screen capture showing the updated Certificate Authorities table.



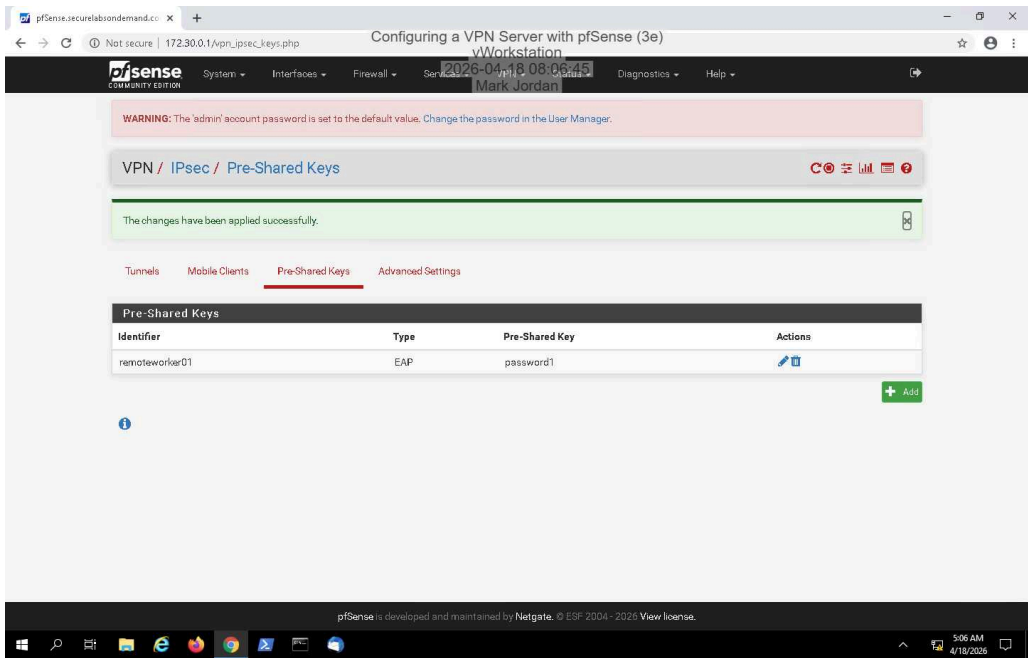
32. Make a screen capture showing the updated Certificates table.



56. Make a screen capture showing the updated IPsec Tunnels table.

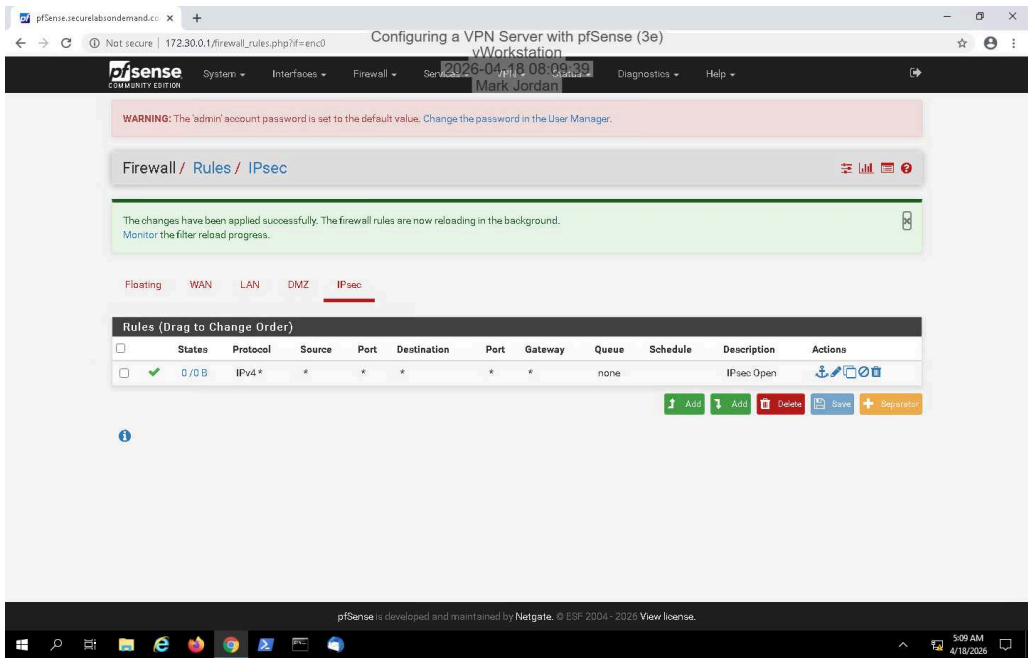


74. Make a screen capture showing the updated Pre-Shared Keys table.



Part 2: Configure a Firewall Rule for VPN Traffic

9. Make a screen capture showing the updated IPsec Rules table.



Section 2: Applied Learning

Part 1: Configure an OpenVPN VPN Server

13. Make a screen capture showing the **CA configuration form**.

pfSense:securelabondemand.co x +

Configuring a VPN Server with pfSense (3e)

Step 5 of 11

2026-04-18 08:13:33

Mark Jordan

Add Certificate Authority

OpenVPN Remote Access Server Setup Wizard

Create a New Certificate Authority (CA) Certificate

Descriptive name OpenVPN_CA_Mark
A name for administrative reference, to identify this certificate. This is the same as common-name field for other Certificates.

Key length 2048 bit
Size of the key which will be generated. The larger the key, the more security it offers, but larger keys take considerably more time to generate, and take slightly longer to validate leading to a slight slowdown in setting up new sessions (not always noticeable). As of 2016, 2048 bit is the minimum and most common selection and 4096 is the maximum in common use. For more information see keylength.com

Lifetime 3650
Lifetime in days. This is commonly set to 3650 (Approximately 10 years.)

Country Code US
Two-letter ISO country code (e.g. US, AU, CA)

State or Province Kentucky
Full State or Province name, not abbreviated (e.g. Kentucky, Indiana, Ontario).

City Shelbyville
City or other Locality name (e.g. Louisville, Indianapolis, Toronto).

Organization Northern Kentucky University
Organization name, often the Company or Group name.

[Add new CA](#)

27. Make a screen capture showing the **Tunnel Settings** section.

pfSense:securelabondemand.co x +

Configuring a VPN Server with pfSense (3e)

Step 6 of 11

2026-04-18 08:19:07

Mark Jordan

Tunnel Settings

Tunnel Network 172.31.1.0/24
This is the virtual network used for private communications between this server and client hosts expressed using CIDR notation (eg. 10.0.8.0/24). The first network address will be assigned to the server virtual interface. The remaining network addresses will be assigned to connecting clients.

Redirect Gateway ☐
Force all client generated traffic through the tunnel.

Local Network 172.30.0.0/24
This is the network that will be accessible from the remote endpoint, expressed as a CIDR range. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.

Concurrent Connections 2
Specify the maximum number of clients allowed to concurrently connect to this server.

Compression Omit Preference (Use OpenVPN Default)
Compress tunnel packets using the LZ0 algorithm. Adaptive compression will dynamically disable compression for a period of time if OpenVPN detects that the data in the packets is not being compressed efficiently.

Type-of-Service ☐
Set the TOS IP header value of tunnel packets to match the encapsulated packet's TOS value.

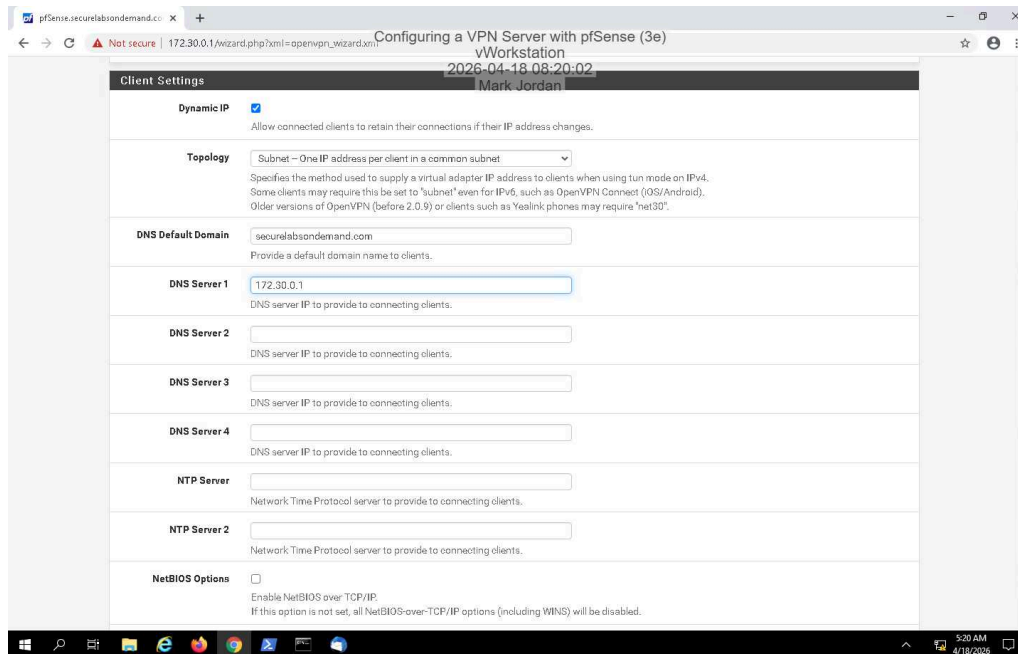
Inter-Client Communication ☒
Allow communication between clients connected to this server.

Duplicate Connections ☐
Allow multiple concurrent connections from clients using the same Common Name.
NOTE: This is not generally recommended, but may be needed for some scenarios.

Client Settings

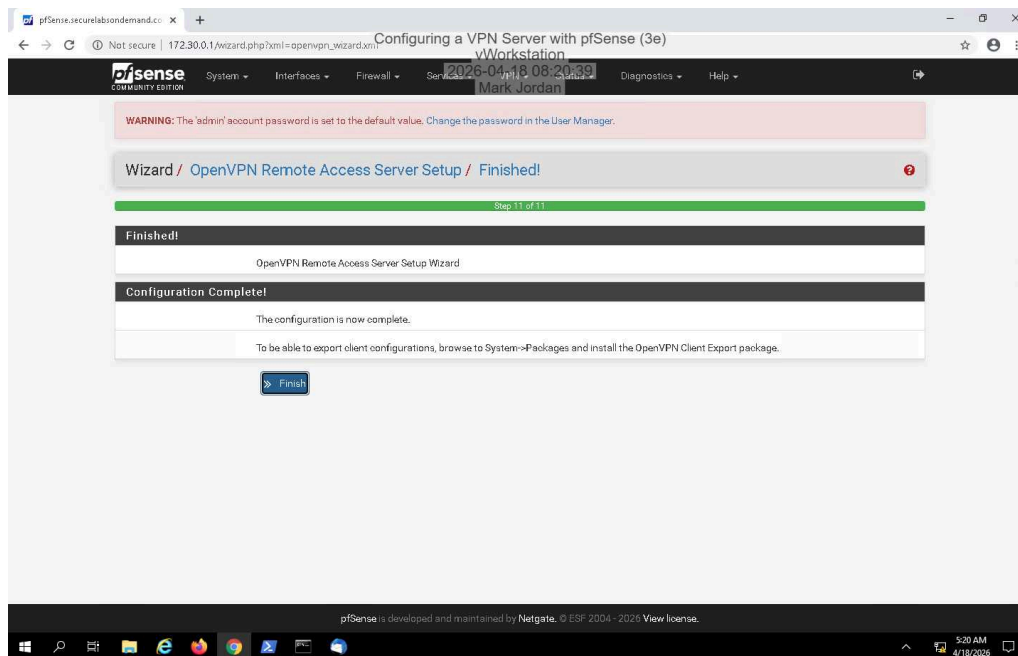
Dynamic IP ☒
Allow connected clients to retain their connections if their IP address changes.

30. Make a screen capture showing the **Client Settings** section.

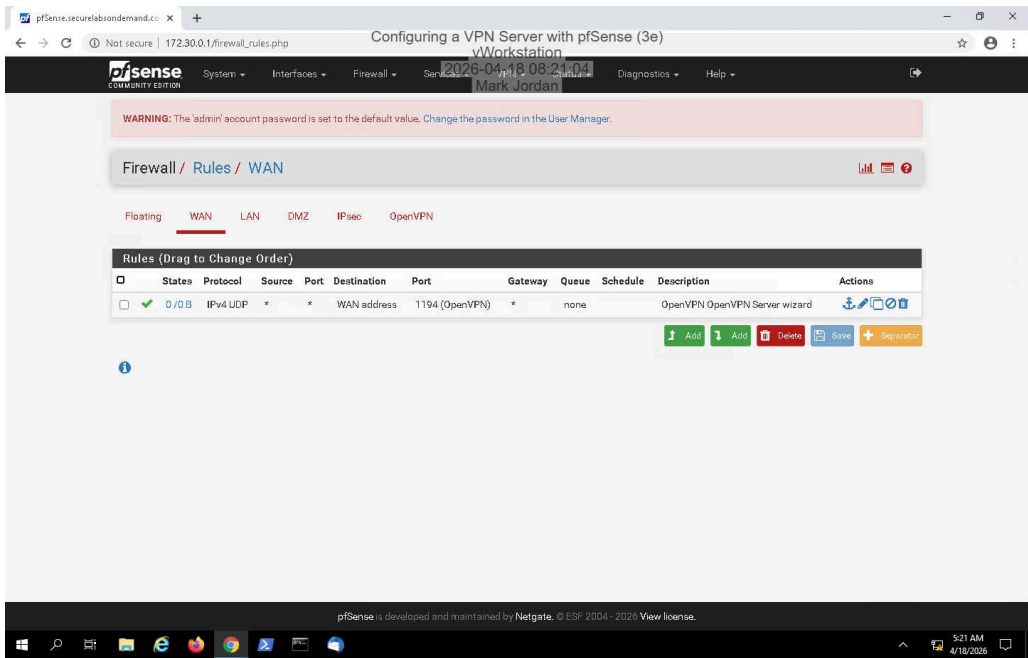


Part 2: Configure a Firewall Rule for VPN Traffic

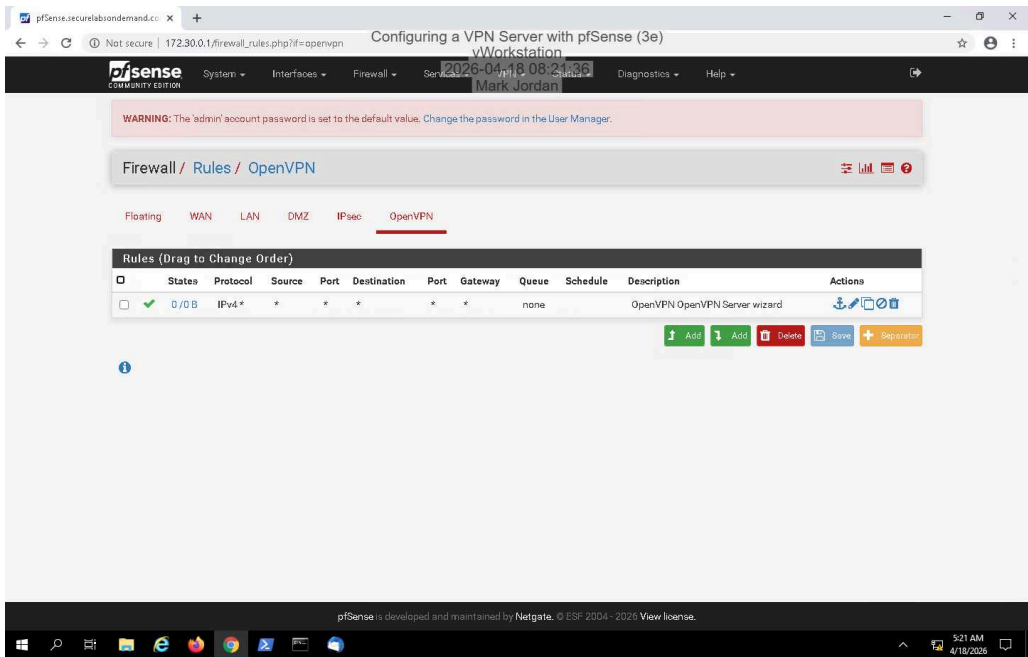
2. Make a screen capture showing the **completed OpenVPN configuration**.



5. Make a screen capture showing the **OpenVPN** rule on the **WAN Rules** table.



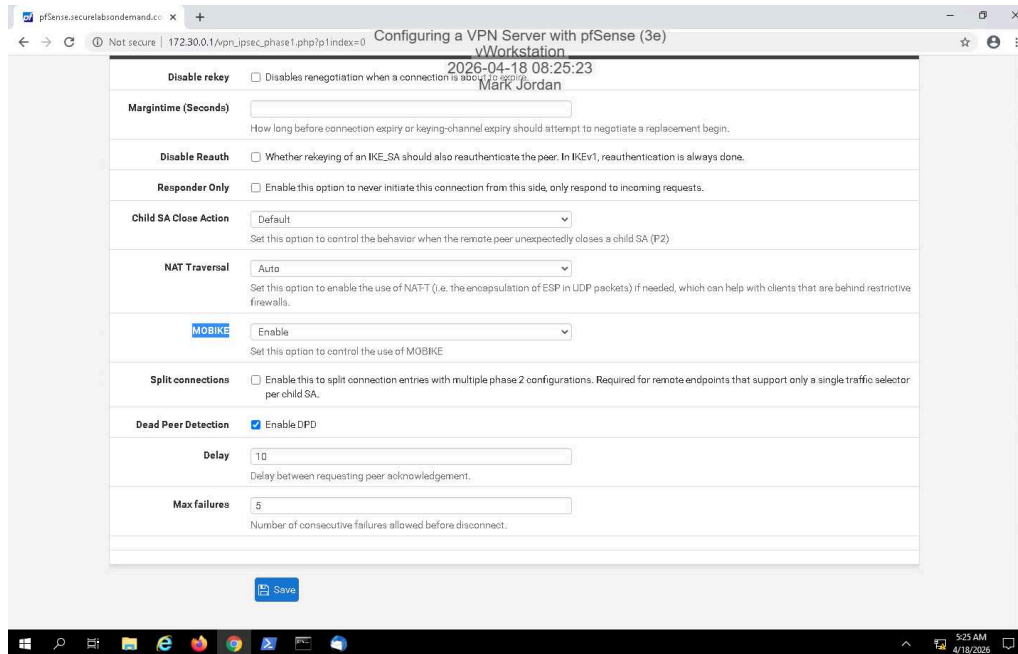
7. Make a screen capture showing the **OpenVPN** rule on the **OpenVPN Rules** table.



Section 3: Challenge and Analysis

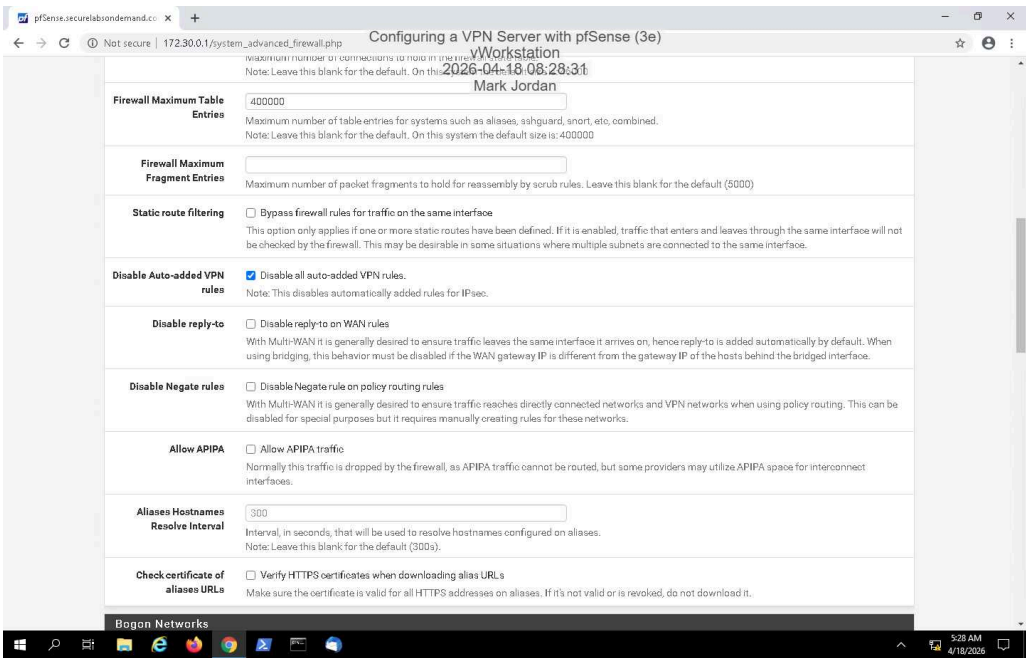
Part 1: Enable IP Roaming for Remote VPN Clients

Make a screen capture showing the enabled **MOBIKE** option in the IPsec tunnel configuration.



Part 2: Create Explicit Firewall Rules for an IPsec VPN

Make a screen capture showing the **disabled automatic IPsec rule creation option**.



Make a screen capture showing your **firewall rules that permit IPsec traffic**.

